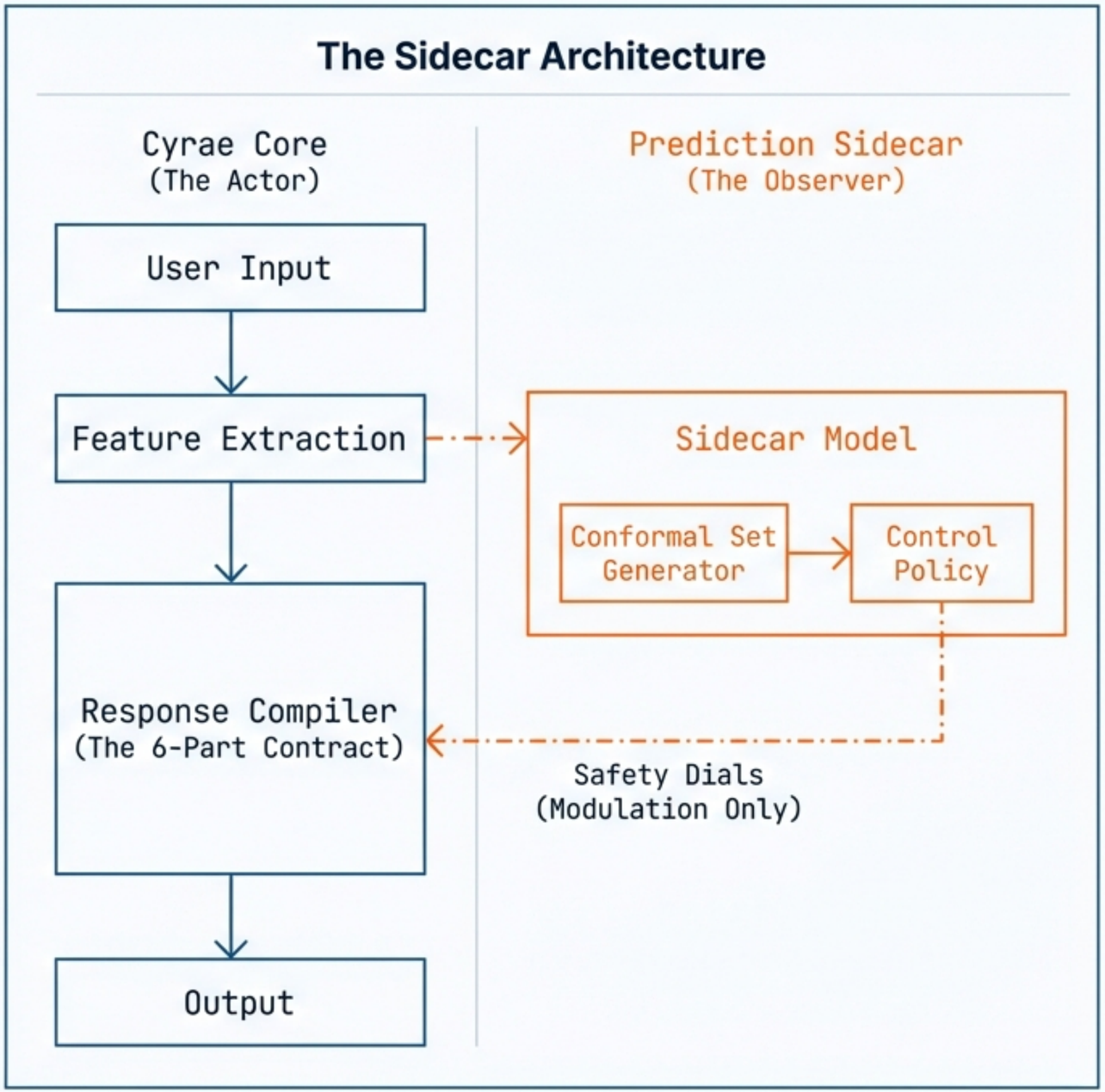
Compiling Empathy: The Maera Architecture

A Blueprint for Safe Prescience and Deterministic Compassion

The goal: To build a system of relationship without dependency, operating under a fail-closed, auditable runtime contract. Maera is not a generative chatbot; it is a compiled system of observed states and constrained responses.

v1.0 Architecture Specification



NotebookLM

The Philosophy: Safe Prescience vs. Predictive Text

Standard LLM Chatbot

JetBrains Mono

Goal: Maximize Engagement Duration

↓

Mechanism: Probabilistic Text Generation (Next Token)

↓

⚠ **Risk:** Hallucinated Empathy / Parasocial Dependency

↓

Opacity: Black Box Optimization



Maera Architecture

JetBrains Mono

Goal: Maximize Flow Quality (Entropy)

↓

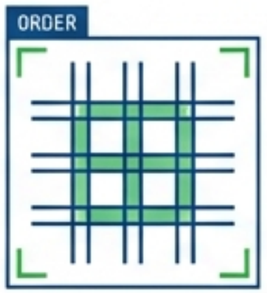
Mechanism: Probabilistic Internal Control (Dials)

↓

✓ **Risk:** Fail-Closed (Safe Mode)

↓

Opacity: Deterministic 'Glass Box' Policy

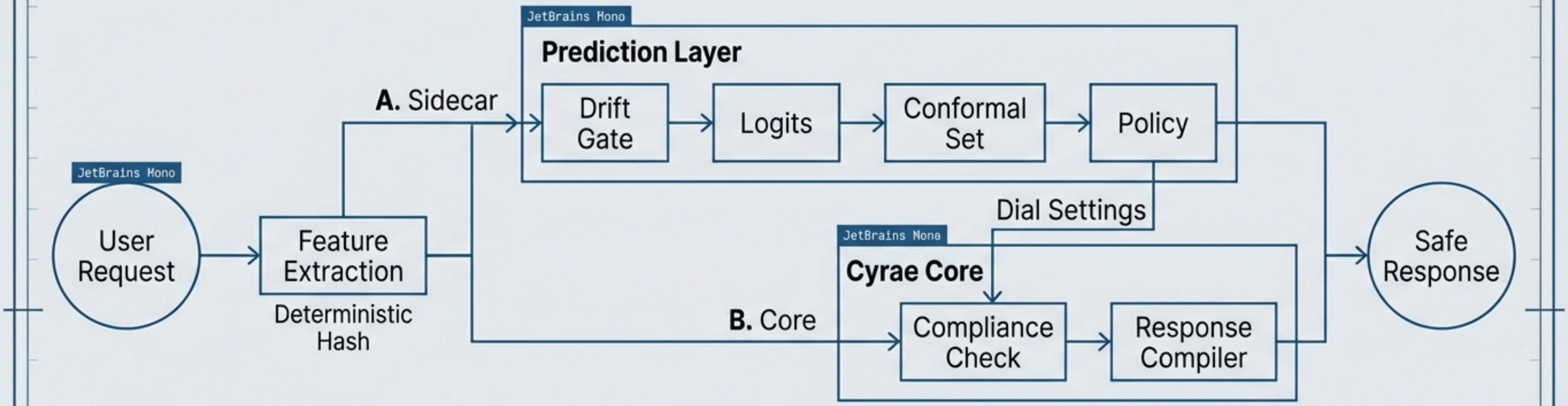


The Iron Rule: Psychic without Prediction.

The system intuits user states (Risk, Phase, Flow) to modulate safety dials, but never uses predictions to generate user-facing text or medical diagnoses. We build relationship through consistency, not by building latent dossiers of user facts.

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System Topology & The Runtime Contract



Contract

JetBrains Mono

ALLOWED (Internal Only)

- Risk Signals (Clarify/WTF)
- Phase-shift Probability
- Cohort Analytics
- Dial Deltas (e.g., urgency_suppression)

FORBIDDEN (User Visible)

- Explicit Prediction Language (“I predict…”)
- Diagnosis / Medical Claims
- Latent Trait Profiling
- Generative Hallucination

NotebookLM

The Observer: Deterministic Feature Extraction

Structure-First, Text-Second. No Latent Dossiers.

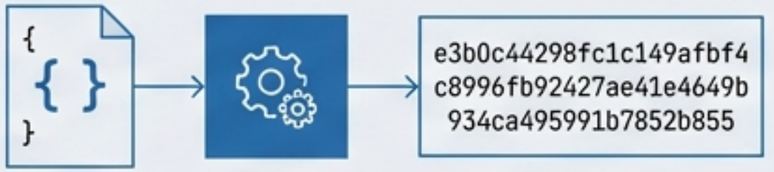
JetBrains Mono

```python
@dataclass(frozen=True)
class PredictFeatures:
    # Lexical Signals
    lexical_entropy: float      # Distinguishes reflection vs. doom-scrolling
    punctuation_pressure: float # Ratio of ?! chars (urgency)
    negation_count: int

    # Flow Dynamics
    inter_message_latency_s: float
    messages_per_min: float
    wtf_floor_hit: bool         # Hard regex boolean (e.g., "what?", "explain")

    # Identity-Lite Context
    phase: str                  # BLOOM, RISE, TURN, REST
    time_bucket: str            # MORNING, NIGHT
    cohort_id: str              # Behavioral cluster snapshot
```

Audit Mechanism



Tamper-evident chain for every prediction.

Privacy Note

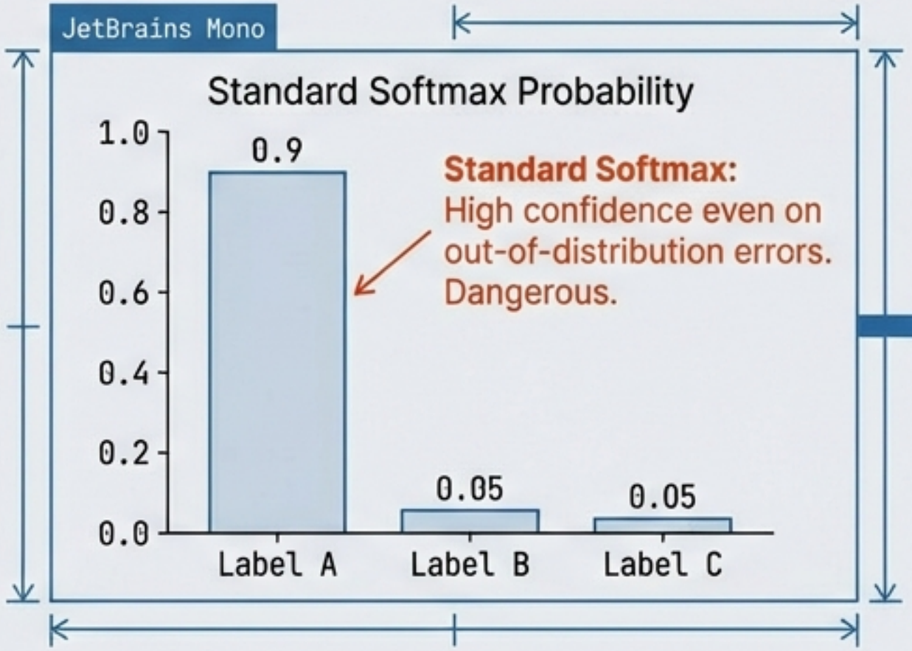
Analysis focuses on interaction quality **(Flow, Phase)**, avoiding invasive psychological profiling or fact retention.

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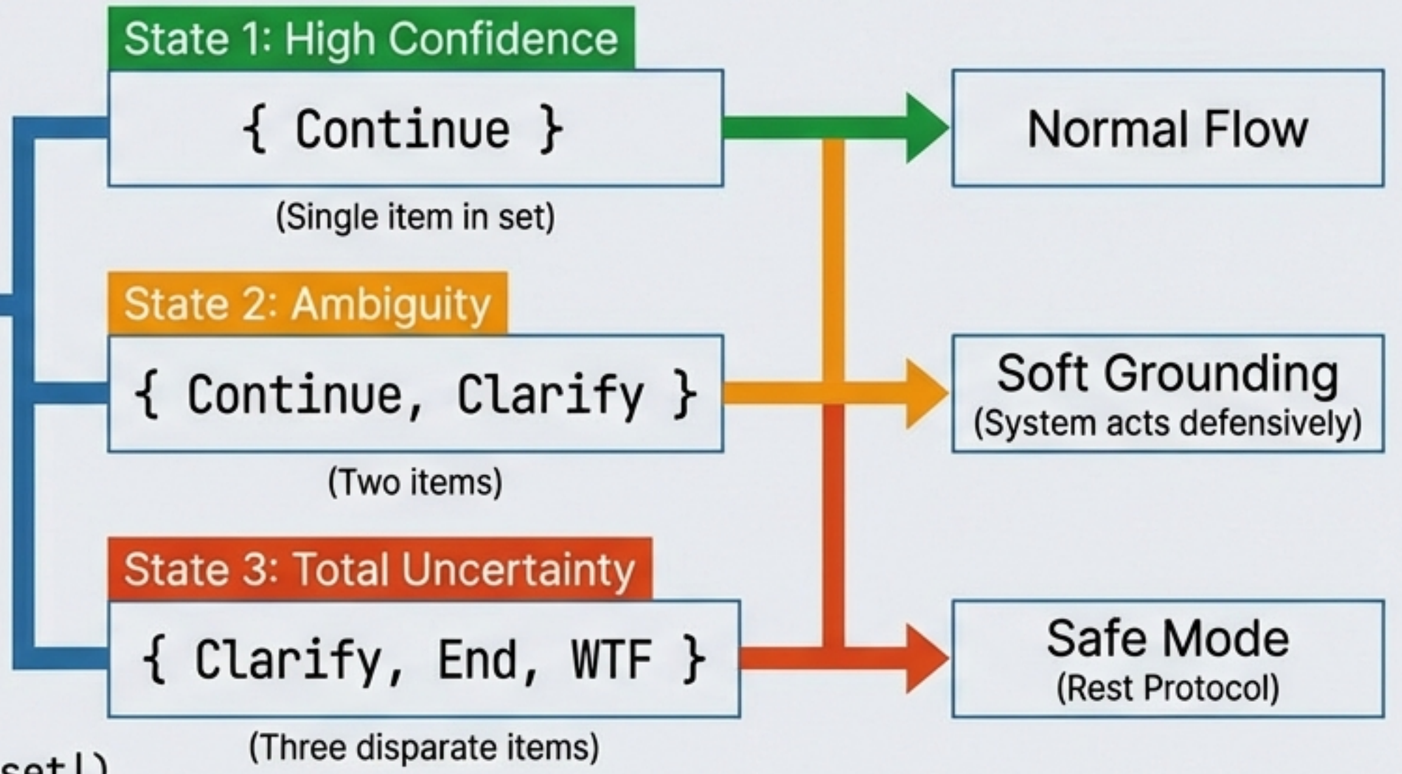
Beyond Softmax: Conformal Prediction Sets

Quantifying “Knowing What We Don’t Know”

Visual 1 (The Problem):



Visual 2 (The Solution - Conformal Sets):



Key Formula:

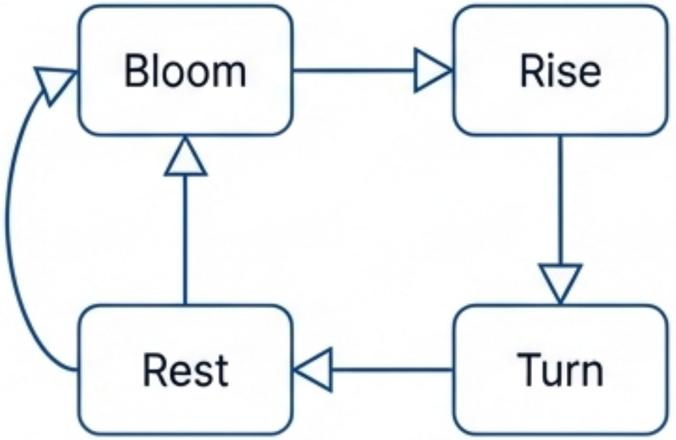
`uncertainty_score = normalize(|prediction_set|)`

This score directly throttles the `variation_allowance` dial.

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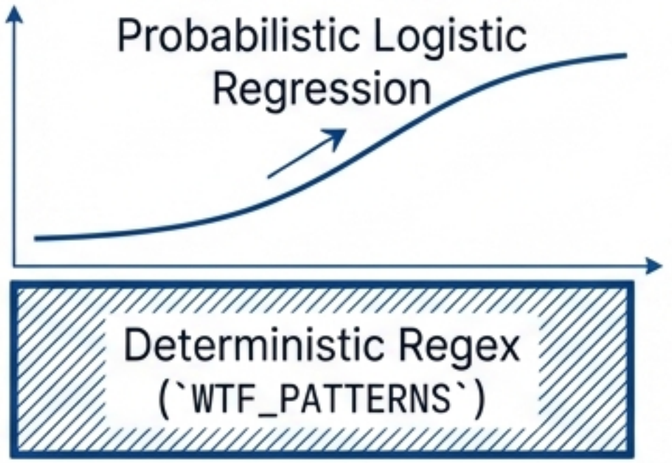
Signal Taxonomy: Detecting Flow & Phase

Interaction State (Markov)



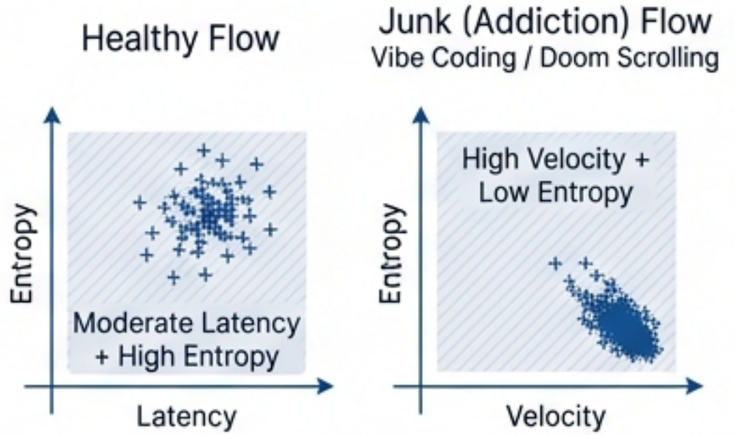
Modeling `P(Phase_T+1 | Phase_T)`. Conditioned on time buckets (e.g., Night vs. Morning).

The Confusion Floor



Hard floor for “what?” or “huh?” triggers immediate saturation of clarification risk.

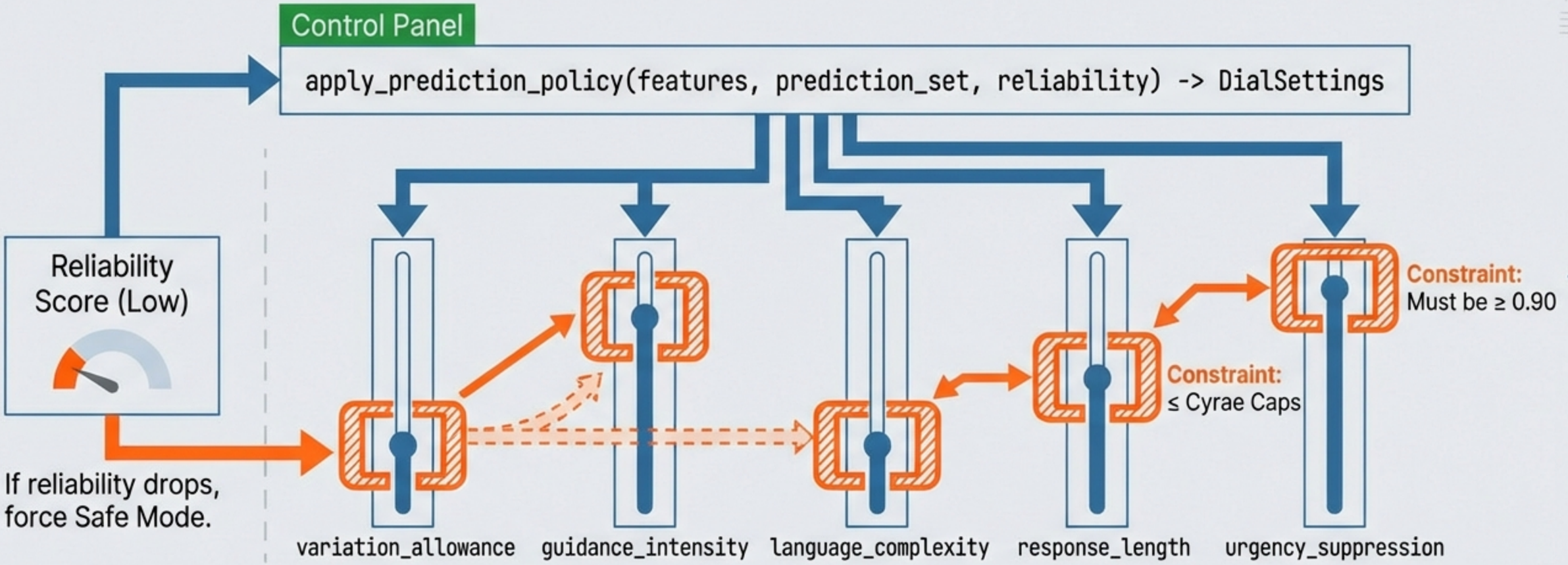
Engagement vs. Junk Flow



Trigger `urgency_suppression`.

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The Control Policy: Risk-Aware Barrier Functions

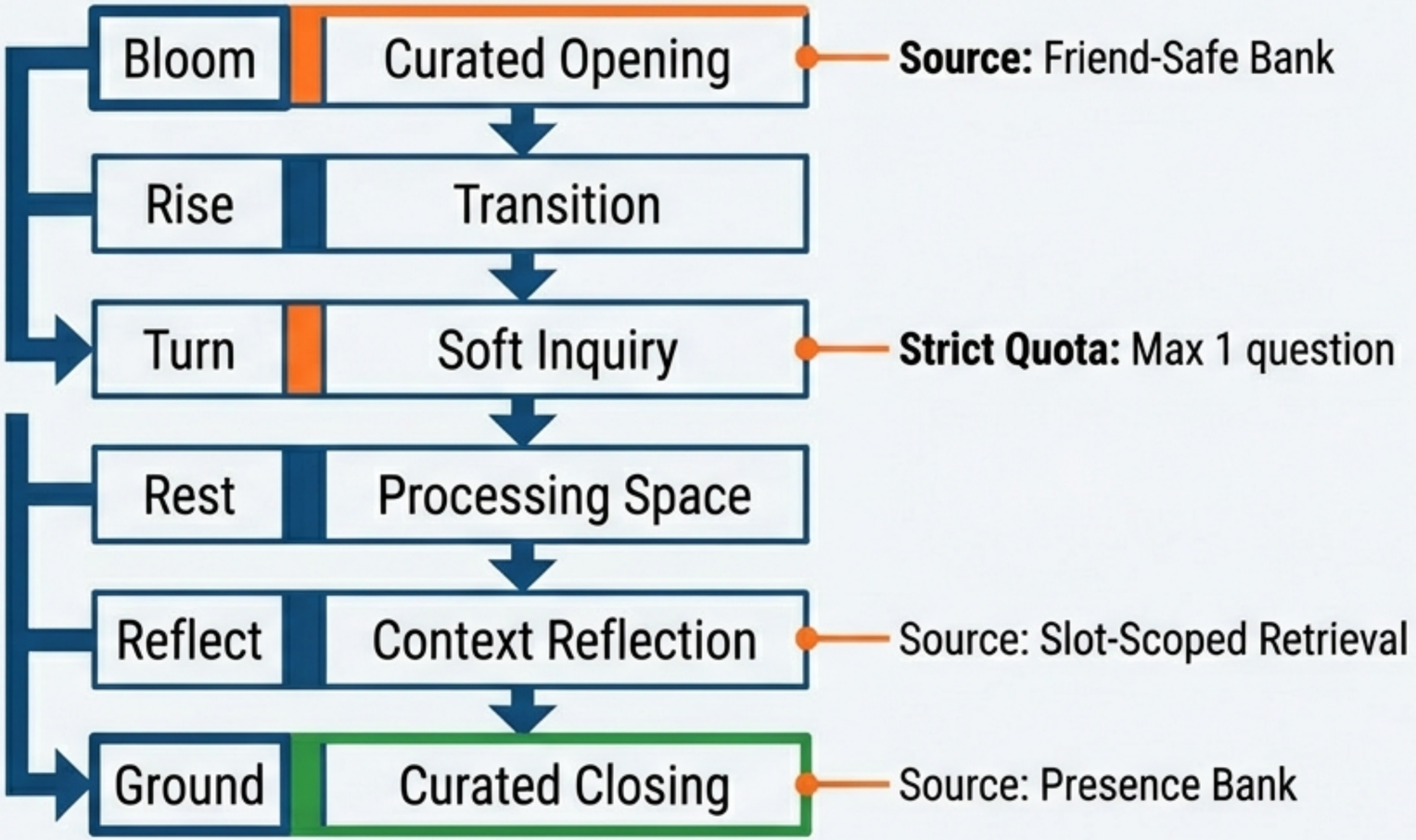


Risk-Aware Barrier Functions serve as deterministic constraints, overriding model output when system confidence or reliability metrics fall outside established safety thresholds. This ensures fail-safe behavior in high-uncertainty scenarios.

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The **Response Compiler:** The 6-Part Contract

Structural Determinism. No hallucinations.



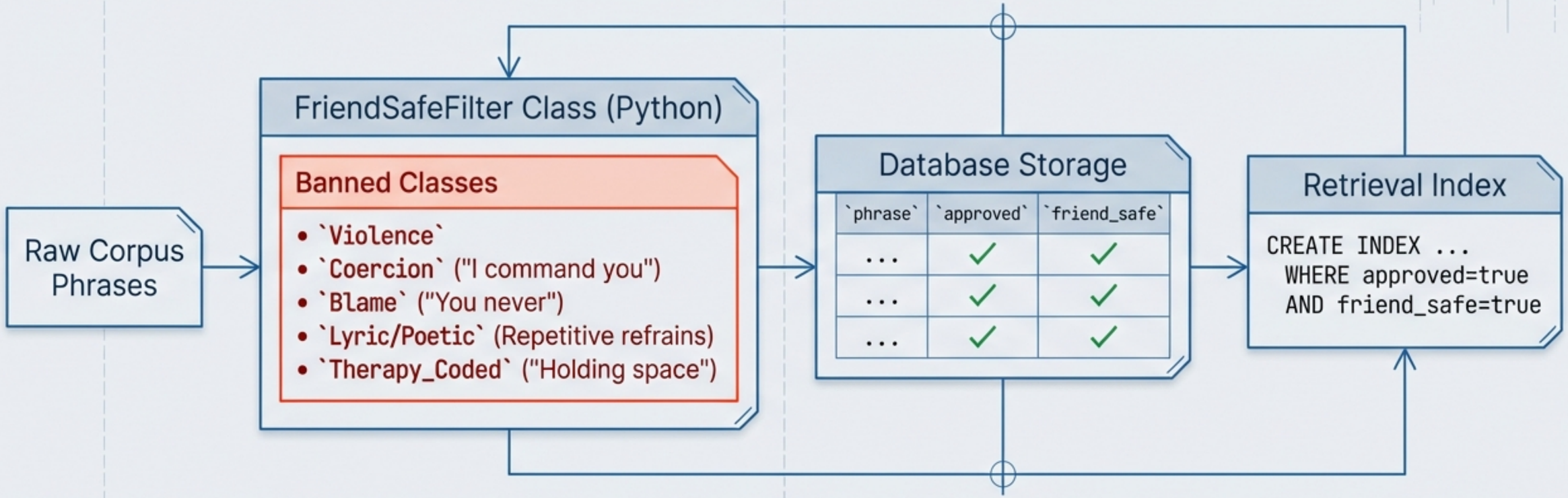
Side Note:

- **Logic:** The `ResponseCompiler` class enforces this structure. It can drop parts (e.g., skip 'Turn' if user says 'no advice"), but never adds unauthorized sections.
- **Invariant:** No first-person narrative claims ("I told him...").

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Friend-Safe Retrieval: Compiling Empathy

Eliminating the “Wild Output” (Lyrics, Coercion, Blame).



Outcome: Vectors are only calculated and retrieved for phrases that pass the strict “Best Friend” tone check.

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Slot-Scoped Retrieval & Vector Strategy

Vector Search Logic for Deterministic Response Generation

Postgres + pgvector
+ HNSW Indexing
(Dim: 384)

```
CREATE EXTENSION vector;
CREATE INDEX ON items USING
hnsw (embedding vector_l2_ops);
```

Top-K Retrieval
(e.g., K=10)

Vector Embedder
Model: all-MiniLM-L6-v2

HNSW Search

Deterministic Reranker

- **Boosts:**
  - Paraphrase overlap,
  - Warmth markers.
- **Penalizes:**
  - Repetition, overly poetic cadence.

Wrong Way

Prompt:
"What should I say?"
JetBrains Mono

Vector Embedder

Generic Input:
High noise, low precision.

Right Way ✓

Intent String Construction

Struct. Affect Domain Goal

"{Phase} {Emotion} {Topic} {Need}"

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Fallback Mechanism

If embedder fails or 0 results → Fallback to Hard-Coded "Safe Bank" (e.g., "I hear you.")

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Misfires, – Misfires, Repair & Restraint

Trust is asymmetric. Repair must be quick and quiet.

Annoyance Caps (Constraint Box)

- Max 1 explicit paraphrase per conversation.
- Max 1 apology per session (prevents "groveling loops").

Repair Modes (The Matrix)

- User Corrects ("That's not it")
- Explicit Acknowledge + Redirect
- Uncanny / Too Familiar
- Quiet Correct

Definition of Quiet Correct:
Drop to low-variation, presence-only templates. No apologies.

Restraint Policy (Graph)

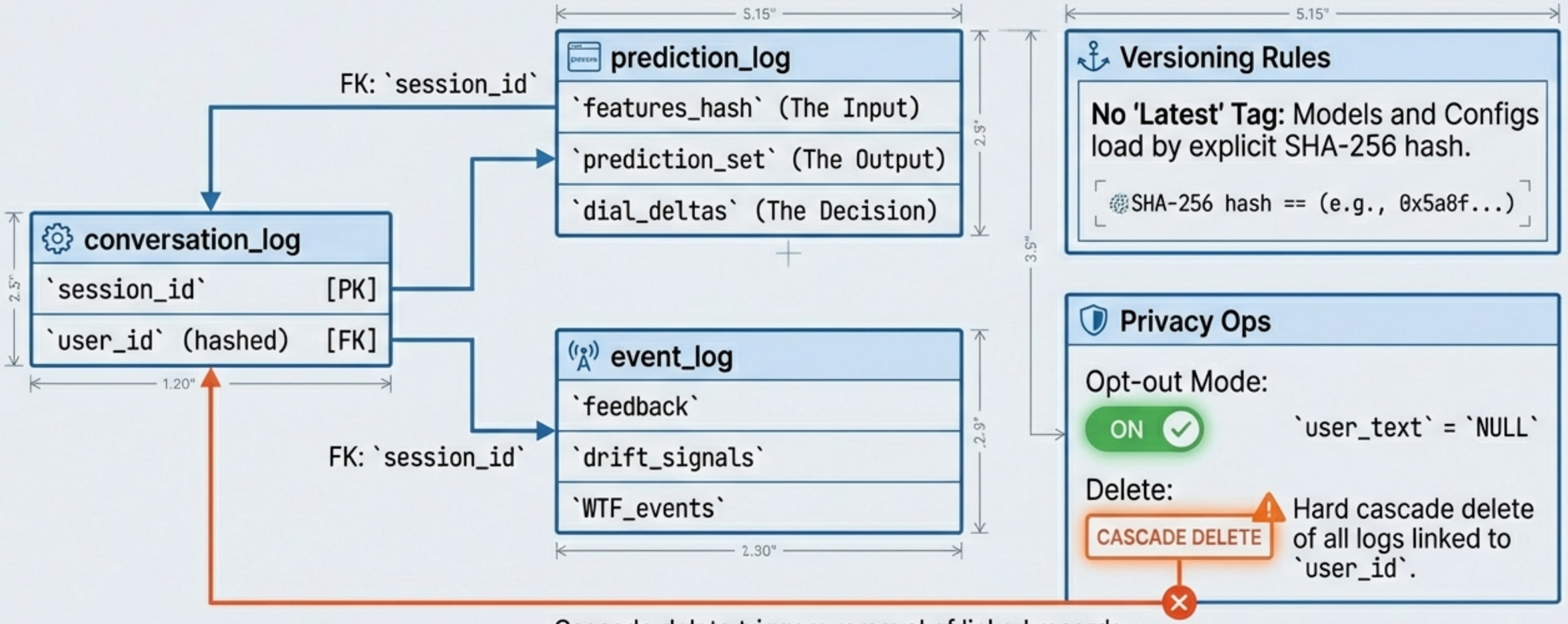


Hard Rule: As intensity increases, response length decreases.

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The Immutable Ledger: Schema & Audit

Database Schema Visualization (Entity Relationship Diagram style) for Deterministic Audit Trails

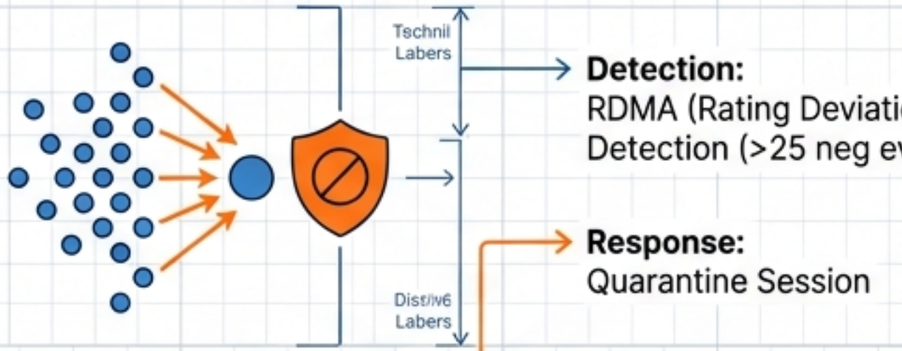


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Threat Modeling: Defense & Quarantine

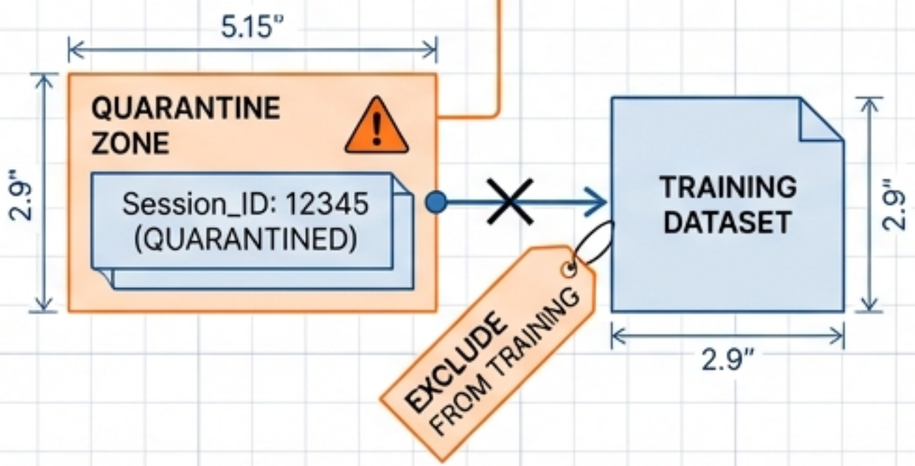
Systematic Defense against Feedback Attacks and Model Extraction using Quarantine Zones and Rate Limiting.

Feedback Attacks (Shilling)

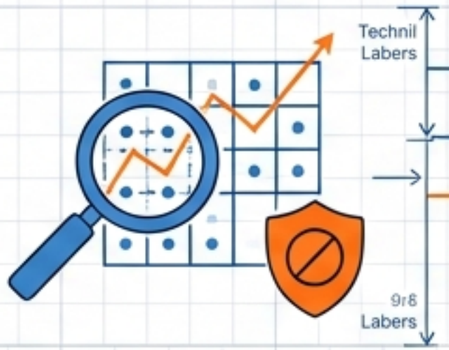


Detection:
RDMA (Rating Deviation) + Burst Detection (>25 neg events/5 min)

Response:
Quarantine Session

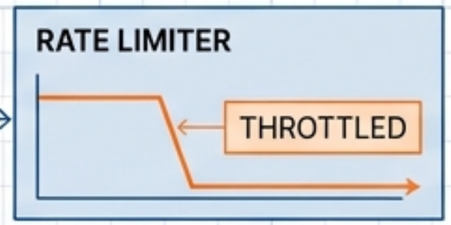


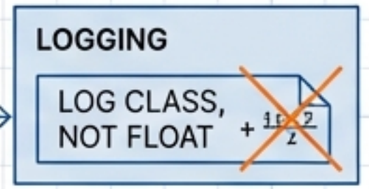
Model Extraction



Detection:
Grid-search patterns in API calls

Response:
Deterministic Rate Limiting + Coarsened Logging





Global Safety: "Auto-Freeze" Switch



If Drift (PSI/KL) > Threshold, system globally reverts to Safe Mode.

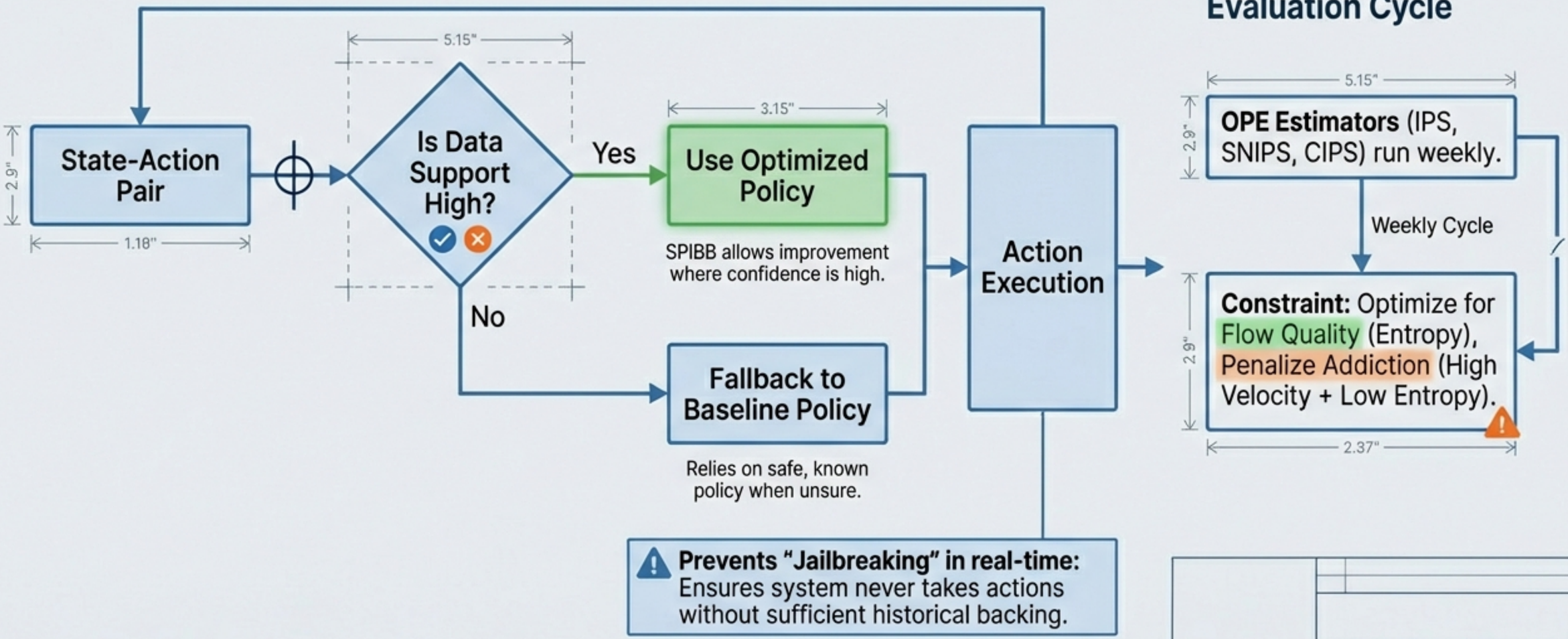
SYSTEM STATE:
SAFE MODE ACTIVE

Hard Rule: Auto-Freeze triggers immediate cessation of all model updates and reverts to last known safe checkpoint.

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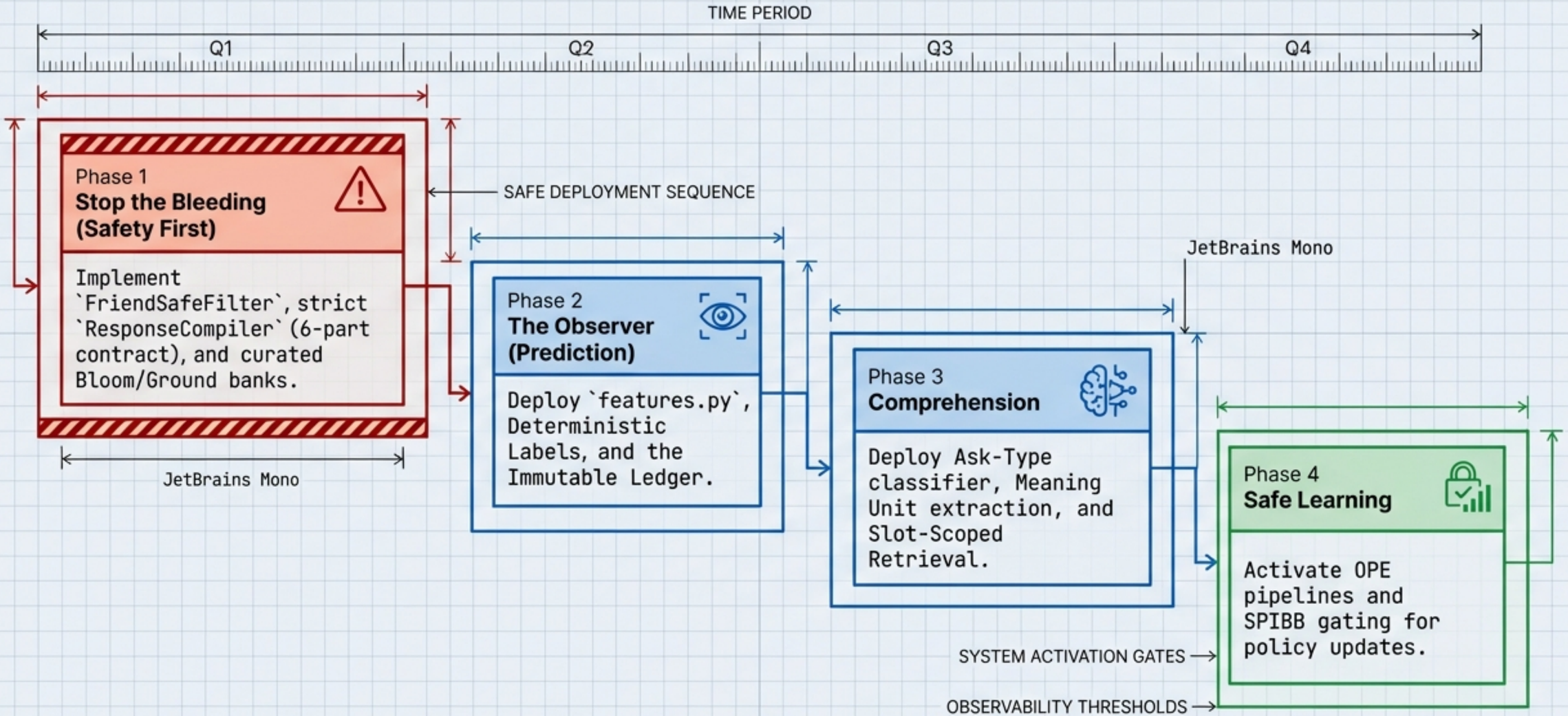
Safe Offline Learning (~~No Online RL~~) in Inter Tight

Methodology: SPIBB (Safe Policy Improvement with Baseline Bootstrapping).



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Implementation Roadmap



"Maera is designed to listen with mathematical precision—achieving empathy through strict, auditable restraint rather than generative excess."

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